Master Electronic Learning Document

Discipline-Specific Knowledge for Intermediate Learners

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Document Description

This document is SSI's comprehensive list of resources on domain-specific knowledge for developing complex satellite systems and general electronic systems and serves as a comprehensive starting place for learning major concentrations of electrical engineering and spacecraft design.

Table of Contents

Power Systems **3**
- DC-DC Converters 3
- Solar Power Generation 3
- Battery Systems 3
- Distribution/Transmission 3
- Thermal Management/Design 3

Analog Systems **4**
- Signal/Power Filtering & Amplification 4
- RF & Wireless Systems 4
- PCB Design & Parasitics 4

Embedded Systems **5**
- Microcontrollers 5
- Digital Architecture 5
- Communication Protocols 5
  - CAN 5
  - I2C 5
  - SPI 5
  - UART 5
  - USB 3.0 5

1-Wire..... 6
Ethernet 802.3..... 6
RS-485 (Physical Layer)..... 6
Sensors & Instrumentation..... 6
Failure Mitigation..... 7
Radiation Tolerance / Hardening..... 7
Shock & Vibration Resistant Design..... 7

Power Systems

DC-DC Converters

- DC/DC Book of Knowledge Practical tips for the User | Mouser
- DC-DC Power Converters
- Introduction to DC/DC Converters
- Lecture 5: Intro to DC/DC, Part 1 | Power Electronics | Electrical Engineering and Computer Science | MIT OpenCourseWare

Solar Power Generation

- Introduction to PV Systems | PVEducation
- Solar Energy - TU Delft OCW

Battery Systems

- Learn About Batteries
- Designing CC-CV Feedback Circuits With the TL103WB
- Battery Charger Fundamentals | Article | MPS
- An Introduction to Batteries: Components, Parameters, Types, and Chargers
- An Introduction to Batteries: Components, Parameters, Types, and Chargers | Article | MPS

Distribution/Transmission

- Principles of Power Integrity for PDN Design—Simplified
- Sitara™ Processor Power Distribution Networks: Implementation and Analysis
- AN-139: Power Supply Layout and EMI | Analog Devices

Thermal Management/Design

- Thermal analysis and control of small satellites in low Earth orbit

- 7.0 Thermal Control - NASA
- INTRODUCTION TO PCB THERMAL ANALYSIS

Analog Systems

Signal/Power Filtering & Amplification

- Signal conditioning - Wikipedia
- An Introduction to Filters - Technical Articles
- Introduction to the Amplifier
- Operational Amplifier Basics - Op-amp Tutorial

RF & Wireless Systems

- Antenna The32ory
    - Antenna Basics
    - List of Antennas (Antenna Types)
- Physical fundamentals of the radar principle - Radartutorial
- Uplink vs Downlink Engineering: Key Differences
- Satellite Uplink and Downlink: How Does it Work? | Bright Ascension
- What Is RF and Why Do We Use It? | Introduction to RF Principles and Components | Electronics Textbook
- Microwaves101 | Basic Concepts

PCB Design & Parasitics

- High Speed Layout Guidelines (Rev. A)
- 10 Best High-Speed PCB Routing Practices | Sierra Circuits
- How Parasitics Impact High-Speed PCB Components
- High-Speed PCB Design Guide
- Parasitics in Precision PCB Layouts
- Guide to Signal Conditioning | Fluke
- Principles of Power Integrity for PDN Design—Simplified
- AN-139: Power Supply Layout and EMI | Analog Devices

Embedded Systems

Microcontrollers

- Introduction To Microcontrollers.pdf
- Introduction to Microcontrollers | ATE Central
- Resources to learn embedded systems engineering. · GitHub

Digital Architecture

- Introduction To Computer System | Beginner's Guide To Computer
- 5.2. The von Neumann Architecture
- Types of Computer Memory and their Applications
- Introduction to Memory and Memory Units - GeeksforGeeks
- Digital Design and Computer Architecture (4th Edition)
- Dive Into Systems | Full Text

Sensors & Instrumentation

- INTRODUCTION TO ELECTRONIC INSTRUMENTATION
- Sensors | Intro to Electrical Engineering | Fiveable
- Understanding Sensor Interfaces - UART, I2C, SPI and CAN
- Sensor Basics: A Beginner's Guide to How Sensors Work, Types, and Uses - AIChipLink
- High Speed Analog to Digital Converter Basics
- High Speed, Digital to Analog Converters Basics

Communication Protocols

- An Advanced Guide to Communication Protocols in Microcontrollers | OPAL-RT
- OSI model - Wikipedia

CAN

- Introduction to the Controller Area Network (CAN) (Rev. B)

I2C

- Application Note - Understanding the I2C Bus
- A Basic Guide to I2C

SPI

- Understanding the SPI Bus

- Basics of SPI: Timing Requirements and Switching Characteristics Serial Peripheral Interface (SPI) for KeyStone Devices User's Guide (Rev. A)

UART

- Universal Asynchronous Receiver/Transmitter (UART) - TI

USB 3.0

- Universal Serial Bus 3.0 (USB 3.0)

1-Wire

- Implementing Enumeration for 1-Wire® Protocol with TMP1826 and TM4C129x Microcontroller

Ethernet 802.3

- A Beginner's Guide to Ethernet 802.3 Application Note (EE-269)

RS-485 (Physical Layer)

- The RS-485 Design Guide (Rev. D)

Failure Mitigation

Radiation Tolerance / Hardening

- What Are Radiation-Tolerant Solutions? | Micron Technology Inc.
- Rad-hard vs. Rad-tolerant: A Complete Guide | VORAGO Technologies
- Radiation Effects on Electronics 101
- Spacecraft electronics design for radiation tolerance
- Space Radiation Effects on Electronic Components in Low-Earth Orbit

Shock & Vibration Resistant Design

- PCB design for extreme environments | NCAB Group
- Vibration analysis of electronic PCB components
- Best Practices for PCB Design in Rugged Environments